WORKED SOLUTION

STEP II 2024, Question 1

Pure • Sums of Consecutive Integers

STEP II 2024, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Both sides are arithmetic series, so each is (number of terms) $\times$ (average term). Equating them gives one quadratic relation between n, k and c, and the rest of the question is number theory on that relation. Fixing k makes it easy to solve for c, and parity or remainders decide when c is a positive integer. Fixing $c = 1$ instead gives an equation of Pell type, and the map in (iv)(b) is the standard way to produce new solutions from old ones.

Solution

Part (i)

The $n + k$ integers from c upwards are $c, c+1, \ldots, c+n+k-1$. Their sum is

$$S_1 = (n+k)c + \frac{(n+k)(n+k-1)}{2}.$$

The next n integers are $c+n+k, \ldots, c+2n+k-1$, with sum

$$S_2 = n(c+n+k) + \frac{n(n-1)}{2}.$$

Doubling both sums and expanding,

$$\begin{aligned} 2S_1 &= 2nc + 2kc + n^2 + 2nk + k^2 - n - k, \\ 2S_2 &= 2nc + 2n^2 + 2nk + n^2 - n. \end{aligned}$$

Subtracting, $2S_1 - 2S_2 = 2kc + k^2 - k - 2n^2$. Hence

$$S_1 = S_2 \iff 2kc + k^2 - k = 2n^2 \iff 2n^2 + k = 2ck + k^2.$$

Every step is reversible (we only expanded, doubled and rearranged), so the equality of sums holds if and only if $2n^2 + k = 2ck + k^2$, as required.

Part (ii)

(a) With $k = 1$ the condition is $2n^2 + 1 = 2c + 1$, so $c = n^2$. This is a positive integer for every n, so *every* positive integer n works, with $c = n^2$. (For example, $n = 2$: $4 + 5 + 6 = 7 + 8$.)

(b) With $k = 2$ the condition is $2n^2 + 2 = 4c + 4$, that is,

$$n^2 = 2c + 1, \qquad c = \frac{n^2 - 1}{2}.$$

For c to be an integer, n^2 must be odd, so n must be odd. For c to be positive we need $n^2 > 1$, which rules out $n = 1$. Conversely every odd $n \geq 3$ gives a positive integer c. So the possible values are

$$n \in \{3, 5, 7, \dots\}, \qquad c = \tfrac{1}{2}(n^2 - 1).$$

The case $n = 3$, $c = 4$ is the equality $4 + 5 + 6 + 7 + 8 = 9 + 10 + 11$.

Part (iii)

With $k = 4$ the condition is $2n^2 + 4 = 8c + 16$, that is,

$$n^2 = 4c + 6 = 4(c + 1) + 2.$$

So n^2 would leave remainder 2 on division by 4. But a square leaves remainder 0 or 1: if $n = 2m$ then $n^2 = 4m^2$, and if $n = 2m + 1$ then $n^2 = 4(m^2 + m) + 1$. So there are no solutions.

Part (iv)

With $c = 1$ the condition becomes $2n^2 + k = 2k + k^2$, that is,

$$2n^2 = k(k + 1). \qquad (*)$$

In other words $n^2 = \frac{1}{2}k(k + 1)$: we want triangular numbers that are also perfect squares.

(a) Trying small k: $k = 1$ gives $2n^2 = 2$, so $n = 1$; then $k = 2, \dots, 7$ give $k(k + 1)/2 = 3, 6, 10, 15, 21, 28$, none of them squares; and $k = 8$ gives $2n^2 = 72$, so $n = 6$. So two possibilities are

$$k = 1,\ n = 1 \qquad \text{and} \qquad k = 8,\ n = 6.$$

(Check the second: $1 + 2 + \cdots + 14 = 105$ and $15 + 16 + \cdots + 20 = 105$.)

(b) Suppose $2N^2 = K^2 + K$. Put $N' = 3N + 2K + 1$ and $K' = 4N + 3K + 1$. Then

$$\begin{aligned} 2N'^2 &= 2(9N^2 + 4K^2 + 1 + 12NK + 6N + 4K) \\ &= 18N^2 + 8K^2 + 24NK + 12N + 8K + 2, \\ K'^2 + K' &= (16N^2 + 9K^2 + 1 + 24NK + 8N + 6K) + (4N + 3K + 1) \\ &= 16N^2 + 9K^2 + 24NK + 12N + 9K + 2. \end{aligned}$$

Subtracting,

$$2N'^2 - (K'^2 + K') = 2N^2 - K^2 - K = 0,$$

using the assumption. So $2N'^2 = K'(K'+1)$. Since N and K are positive integers, so are N' and K', and by $(*)$ and part (i) the value N' is a possible value of n, with corresponding $k = K'$.

Ask yourself Where do these strange formulas come from?

Multiply $(*)$ by 4 and add 1: it becomes $(2k+1)^2 - 8n^2 = 1$. With $x = 2k+1$ and $y = n$ this is the Pell equation $x^2 - 8y^2 = 1$, whose solutions are generated by multiplying $x + y\sqrt{8}$ by the smallest solution $3 + \sqrt{8}$. Writing $(x + y\sqrt{8})(3 + \sqrt{8}) = (3x + 8y) + (x + 3y)\sqrt{8}$ and translating back to k and n gives exactly $N' = 3N + 2K + 1$ and $2K' + 1 = 3(2K+1) + 8N$, i.e. $K' = 4N + 3K + 1$.

(c) Applying the map to $(N, K) = (1, 1)$ just gives $N' = 6$, $K' = 8$, the second solution from (a). Applying it to $(N, K) = (6, 8)$:

$$N' = 18 + 16 + 1 = 35, \qquad K' = 24 + 24 + 1 = 49,$$

and applying it again to $(35, 49)$:

$$N' = 105 + 98 + 1 = 204, \qquad K' = 140 + 147 + 1 = 288.$$

So two further solutions are

$$k = 49,\ n = 35 \qquad \text{and} \qquad k = 288,\ n = 204.$$

Indeed $2 \cdot 35^2 = 2450 = 49 \cdot 50$ and $2 \cdot 204^2 = 83232 = 288 \cdot 289$.

Checks

- The opening example has $n = 3$, $k = 2$, $c = 4$: $2 \cdot 9 + 2 = 20 = 2 \cdot 4 \cdot 2 + 4$, so it satisfies the relation in (i), and fits (ii)(b) with $c = \frac{1}{2}(9 - 1)$.
- A computer search over all $n, k < 40$ and $c < 200$ confirmed that the two sums agree exactly when $2n^2 + k = 2ck + k^2$, and a search with $c = 1$ and $n, k < 400$ found only $(k, n) = (1, 1), (8, 6), (49, 35), (288, 204)$, matching (iv).
- In (iv)(b), note that the step from (N, K) to (N', K') preserves the quantity $2N^2 - K^2 - K$ for *any* N, K, which is a quick way to test your algebra with small numbers, say $N = K = 0$: then $N' = K' = 1$ and both values are 0.

Common slips

- Including $n = 1$ in (ii)(b): it gives $c = 0$, which is not a positive integer.
- In (i), miscounting where the second block starts: it begins at $c + n + k$, not $c + n + k + 1$.

- In (iv)(c), giving $k = 8$, $n = 6$ again, which is the first output of the map but was already found in (a).

Beyond STEP

The equation $x^2 - 8y^2 = 1$ is a Pell equation, and for any non-square d the solutions of $x^2 - dy^2 = 1$ all come from powers of a single fundamental solution, through multiplication of numbers $x + y\sqrt{d}$. This is the first glimpse of units in rings like $\mathbb{Z}[\sqrt{d}]$, a central topic in algebraic number theory. Davenport's *The Higher Arithmetic* treats Pell's equation through continued fractions.

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